Where Literacy Moderates Momentum: A Characterization Across Institutional Ownership Terciles, U.S. Equities 2009–2023

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Abstract

On the stable-headquarters subsample of CRSP U.S. equities 2009–2023, the three-way of momentum, idiosyncratic volatility (IV), and state National Financial Capability Study (NFCS) Big-Five literacy concentrates in the mid-tercile of institutional ownership (IO): $\hat{\gamma}_{\text{mid}} = -0.0321$, wild-cluster $p = 0.064$. Literacy dominates a state-college-attainment horse-race within mid-IO. The mid-IO coefficient collapses 26-fold and the prior aggregate three-way prediction collapses 39-fold under a point-in-time (PIT) headquarters correction; the within-mid-IO IV-decile decomposition fails to reject decile equality, leaving com-

positional heterogeneity indistinguishable from a no-channel null.

1 Introduction

This paper characterizes where in the cross-section of institutional ownership (IO) the literacy moderation of the momentum–idiosyncratic-volatility (IV) interaction concentrates. The headline finding is functional-form-free. On the stable-headquarters subsample of CRSP-listed U.S. common stock 2009–2023 (519,090 firm-months, 6,081 firms), stratifying the three-way regression by within-stock IO tercile concentrates the literacy moderation in the mid-IO tercile: $\hat{\gamma}_{\text{mid}} = -0.0321$ (state-clustered $t = -2.63$; wild-cluster bootstrap $p = 0.064$; family-wise Romano-Wolf $p = 0.077$). The low- and high-IO coefficients are one-third and one-sixth of the mid-tercile magnitude. A continuous-margin quadratic-IO specification corroborates the non-monotone shape with an interior trough at $\text{IO}^* \approx 0.34$ ($\hat{\gamma}_5 = +0.0151$, $p = 0.017$). Literacy dominates the closest education proxy: a state-college-attainment horse-race within mid-IO returns $\hat{\gamma}_{\text{lit}} = -0.0350$ ($t = -2.97$) and $\hat{\gamma}_{\text{college}} = +0.00076$ ($t = +0.65$).

The contribution is the discrete-tercile mid-IO concentration on stable-headquarters and the disciplining of the within-family interpretation against the alternatives the design separates and those it does not. The institutional setting is the one in which retail investors overweight geographically proximate firms [Coval and Moskowitz, 1999, Ivković and Weisbenner, 2005] and local investor-base concentration moves prices through a demand channel [Hong et al., 2008]; geographic information asymmetry also shapes institutional behavior [Bernile et al., 2015]. The operative friction is the interaction of limits to arbitrage that scale with idiosyncratic volatility [Shleifer and Vishny, 1997, Pontiff, 2006] with sophistication-dependent local retail demand modulated by state National Financial Capability Study (NFCS) Big-Five literacy [Lusardi and Mitchell, 2014, Calvet et al., 2009, Grinblatt et al., 2012].

The within-family interpretation is held tightly. The mid-IO concentration discriminates against any generic state-level moderator that hits all headquarters-state stocks alike, but it does not select a single within-family microfoundation. Within mid-IO, the headline three-way decomposed by within-month NYSE IV decile fails to reject equality across deciles (joint χ^2 $p = 0.94$ persistent; $p = 0.18$ time-varying), and decile 10 (where a canonical limits-to-arbitrage arbitrage-capacity reading predicts the strongest moderation) returns $\hat{\gamma} \approx 0$. The data therefore rule out the canonical arbitrage-capacity microfoundation, but the IV-decile null cannot distinguish compositional heterogeneity from a no-channel null. I name compositional heterogeneity as the most direct family-internal description of what remains rather than as a positively-identified channel. The quadratic-IO curvature stability across the 2015 break is uninformative under a Chow $\chi^2(1)\, p = 0.997$ where the contrast standard error is approximately ninety percent of $\hat{\gamma}_5$ itself; the trough IO* drifts from 0.26 to 0.40 as the median IO share rises (Section 5.2).

Three bounds discipline the result. First, the quadratic five-percent statistical clearance is the central inferential anchor for the parametric corroboration but does not survive log-IO, kink-IO, or a twenty-bin diagnostic; the discrete-tercile contrast is the functional-form-free characterization, and the quadratic provides parametric corroboration only. Second, the mid-IO coefficient collapses 26-fold ($-0.0321 \rightarrow -0.00121$) under a point-in-time (PIT) EDGAR 10-K-header literacy reassignment that captures the 14 percent of firm-months whose then-current headquarters state differs from the static Compustat snapshot, and the aggregate three-way collapses approximately 39-fold ($-0.0117 \rightarrow -0.0003$), refuting the prior aggregate-three-way prediction; the within-stock retail-versus-institutional difference also sign-flips. The directional within-tercile pattern is preserved on the PIT full panel; the magnitude is not. Third, on the stable-HQ subsample the mid/low ratio is 2.83 versus a full-panel ratio of 1.09; relocators are nearly twice as concentrated in the low-IO tercile as elsewhere, and dropping them attenuates low-IO. About one-third of the gap comes from low-IO attenuation and two-thirds from mid-IO growth, so the underlying within-stock contrast on the corrected full panel is smaller than the stable-HQ ratio implies. The corrected sample refutes the prior aggregate prediction; the documented mid-IO concentration on stable-HQ replaces it as the contribution.

The closest prior work is Korniotis and Kumar [2013], who show that state-level business cycles predict returns of locally headquartered firms. I differ on three margins: the state variable is NFCS literacy on a three-year wave frequency rather than the quarterly business-cycle frequency; the structural form is a three-way moderation of an anomaly interaction; and the discriminating margin is a within-stock IO contrast that the state business-cycle channel does not predict. A business-cycle control block attenuates the aggregate literacy three-way by 20–23 percent, so the two moderators are distinct but partly overlapping; the discrimination claim's scope is generic state-level confounds, not ownership-interactive ones, which remain open. Korniotis et al. [2020] use brokerage-demographic proxies for state-level sophistication; that paper uses neither NFCS literacy nor the momentum-IV interaction. Hillert et al. [2014] is the closest precedent for a state-level moderator of momentum, with media coverage as the primary moderator and state investor individualism as a cross-sectional moderator of the media-momentum effect, rather than literacy.

2 Data and Sample Construction

2.1 Sample

The sample is the cross-section of CRSP-listed common stock (share codes 10 and 11; exchange codes 1, 2, 3, covering NYSE, NYSE American, and NASDAQ) from January 2009 through December 2023. The full panel contains 623,896 firm-month observations, 7,447 unique firms, 51 headquarters states (the fifty states plus the District of Columbia), and 180 calendar months. I z-score all focal variables within each calendar month, so the residual variation the regressions exploit is cross-sectional and free of time-aggregate movements. State financial literacy is measured from the NFCS Big Five literacy questions, fielded by the FINRA Investor Education Foundation in waves 2009, 2012, 2015, 2018, and 2021, and forward-filled to monthly frequency between waves.

2.2 The stable-headquarters subsample

The local-bias channel transmits through community ties between a firm and its headquarters-state retail investor base. Such ties take time to form and to decay, so for firms that relocate headquarters within sample the effective local-investor population is ambiguous by construction. I construct a stable-headquarters subsample that drops firms with a confirmed in-sample EDGAR-based relocation, using a union of two flags: `hdr_changed` fires when the 10-K-header business-address `STATE` changes across a firm's in-sample 10-Ks (954 firms); `sec_disagree` fires when the firm's most-recent 10-K-header state disagrees with the static Compustat snapshot and the firm has at least two parsed 10-K headers (524 additional firms). The union identifies 1,366 relocator firms (18.9 percent of flag-covered firms, 16.8 percent of firm-months). The top first-to-last state pairs are consistent with the headquarters-relocation mechanism of Pirinsky and Wang [2006], who show that relocating firms shift their local return comovement from origin to destination state. The stable-headquarters subsample contains 519,090 firm-months and 6,081 firms.

The paper's headline patterns are estimated on the stable-headquarters subsample under static-snapshot literacy; the point-in-time literacy reassignment on the full panel is the correctness check (Section 4.6). Construction detail and PIT-coverage diagnostics are in IA A and IA B.

2.3 Variable definitions

The dependent variable is the monthly stock return `ret` from CRSP. Momentum is `mom_12_2`, the standard Jegadeesh and Titman [1993] cumulative return from twelve months before to two months before the observation month, skipping the most recent month. Idiosyncratic volatility (IV) is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. State financial literacy is the NFCS Big Five share-correct score, z-scored within month and denoted `literacy_z`.[1] Each cross-product (the two-way terms mom $\times$ IV, mom $\times$ `literacy_z`, IV $\times$ `literacy_z`, and the three-way mom $\times$ IV $\times$ `literacy_z`) is z-scored within month. The Fama-French 12-industry block and the Korniotis and Kumar [2013] business-cycle block enter robustness specifications.

[1] Correct-answer keys M6=1, M7=3, M8=2, M9=2, M10=1 (FINRA codebook; full construction detail in Appendix A). Correct-answer keys are verified from the FINRA Foundation NFCS State-by-State Codebook, 2021 version, available at finrafoundation.org/knowledge-we-gain-share/nfcs/data-and-downloads.

2.4 Institutional ownership from the Thomson Reuters s34 13F panel

I build a firm-quarter institutional-ownership (IO) panel from the WRDS Thomson Reuters s34 13F holdings file for 2009Q1 through 2023Q4 (222,180 firm-quarter observations, 7,907 distinct permnos). The numerator is constructed under the literature-standard convention: amendment-deduplicated filings (keep only the latest filing per (manager, security, report-date) triple) and the sole-plus-shared investment-discretion measure, rather than the raw `shares` field. Under this construction, the residual above-one rate is 0.36 percent (vs 6.49 percent under raw `shares`); I cap the residual at one. Full construction detail, with the amendment-deduplication rule keyed on `fdate` via `DISTINCT ON (mgrno, cusip) ORDER BY fdate DESC`, is in IA P.

The persistent IO measure is each firm's time-mean institutional-ownership share across its available Thomson quarters; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier (a strictly conservative no-look-ahead merge). On the stable-headquarters subsample, persistent IO covers 518,962 firm-months and time-varying IO covers 493,889 firm-months.

2.5 Descriptive statistics

Table 1 reports the cross-sectional distribution of the focal variables.

3 Identification

3.1 Design and variation exploited

The primary design is a cross-sectional two-way fixed-effects (TWFE) panel regression with state and month fixed effects. The variation it exploits is the cross-state, within-state, and cross-NFCS-wave variation in `literacy_z`, interacted with the within-month cross-section of momentum and idiosyncratic volatility (IV). State fixed effects absorb time-invariant state characteristics; month fixed effects absorb aggregate time-varying shocks. The residual variation that identifies the three-way coefficient is the differential within-state-month sorting of mom $\times$ IV across firms whose headquarters-state literacy differs. The headline specification is

$$\begin{aligned} r_{i,s,t} = \alpha_s + \tau_t + \gamma \left(\mathrm{mom}_{i,t} \cdot \mathrm{IV}_{i,t} \cdot \texttt{literacy\_z}_{s(i,t),t}\right) \\ + \delta_1 \left(\mathrm{mom}_{i,t} \cdot \mathrm{IV}_{i,t}\right) + \delta_2 \left(\mathrm{mom}_{i,t} \cdot \texttt{literacy\_z}_{s(i,t),t}\right) + \delta_3 \left(\mathrm{IV}_{i,t} \cdot \texttt{literacy\_z}_{s(i,t),t}\right) \\ + \boldsymbol{\beta}'\mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}, \end{aligned} \tag{1}$$

where $r_{i,s,t}$ is the month-t return of firm i headquartered in state $s(i,t)$, α_s and τ_t are state and month fixed effects, $\mathbf{X}_{i,s,t}$ collects the lower-order terms and (in robustness specifications) the FF12 industry block, the Korniotis and Kumar [2013] business-cycle block, and the college-attainment interaction block. The state-assignment function $s(i,t)$ distinguishes two merge conventions: under the static-snapshot Compustat merge, $s(i,t) = s_i$ for all t; under the point-in-time (PIT) EDGAR 10-K-header reassignment, $s(i,t) = s_i(t)$ varies within firm. On the stable-HQ subsample, $s_i(t) = s_i$ for all in-sample t and the two merges coincide. Equation (1) is an estimating equation, not a derived structural relation; γ is the conditional association the design recovers. The continuous-margin quadratic-IO specification of Section 4.2 (Equation 2) augments Equation (1) with the four-way IO and IO^2 interactions and all 21 lower-order terms. The discrete-tercile rendering re-estimates Equation (1) separately within each within-stock IO tercile.

3.2 Identifying assumptions and their diagnostics

Conditional unconfoundedness of `literacy_z` for the three-way. Conditional on state and month fixed effects, the lower-order interactions, and the controls in $\mathbf{X}$, residual variation in state-aggregate `literacy_z` is uncorrelated with omitted return determinants that interact with mom $\times$ IV in the same functional shape. The paper addresses the assumption three ways: by absorbing the FF12 industry block; by running a state college-attainment horse-race (literacy dominates within mid-IO, $\hat{\gamma}_{\text{lit}} = -0.0350$ vs $\hat{\gamma}_{\text{college}} = +0.00076$; Section 4.7); and by building a within-stock institutional-ownership contrast that no generic state-level moderator can fake. Section 4.1 reports the within-stock contrast. The discrimination claim's scope is generic state-level confounds that hit all headquarters-state stocks alike; ownership-interactive state-level confounds remain an open class (Section 7).

Inference with 51 state clusters. The natural clustering dimension is the state, and with 51 clusters asymptotic cluster-robust standard errors over-reject. The pre-registered design named two-way state-by-month CGM clustering as the headline; on inspection two-way CGM over-states the evidence by absorbing time-cluster correlation that is mechanically zero after within-month z-scoring (see Section 6.4), so the binding inference is the one-way state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed; the two-way state-by-month CGM is reported as a diagnostic in IA I.

Headquarters-state assignment and the literacy-merge convention. The static Compustat snapshot mis-assigns relocator firm-months, so the static-snapshot literacy merge on the full panel contaminates the literacy variable for the 14 percent of firm-months whose PIT headquarters differs from the static one. The paper addresses the contamination two ways: headline patterns are estimated on the stable-headquarters subsample, on which the static and PIT merges coincide and the local-investor unit of analysis is held fixed; the PIT literacy reassignment on the full panel is the strictest available correction, reported as the primary correctness check (Section 4.6).

3.3 The estimand

Under the conditional-unconfoundedness assumption, the TWFE state-plus-month $\hat{\gamma}$ is the associational three-way cross-sectional coefficient: the average within-state-month differential sensitivity of returns to mom $\times$ IV as a function of state-aggregate `literacy_z`, partialing out state and month means. It is not a clean average treatment effect of literacy. Causal language is bounded accordingly.

IA T reports a staggered-adoption difference-in-differences design around state personal-finance high-school mandates as a sign-direction sanity check, not as identifying variation that supports the cross-sectional headline. The cross-sectional contribution stands independent of the DiD.

4 Results

The headline finding is functional-form-free: stratifying the three-way regression by within-stock IO tercile concentrates the literacy moderation in the mid-IO tercile of the stable-headquarters subsample. I present this discrete-tercile contrast first, then the corroborating continuous-margin quadratic specification, the relocator decomposition that bounds the sample-selection contribution to the mid/low ratio, and the pre/post-2015 split. Section 6 reports the functional-form grid that disciplines the quadratic specification's five-percent clearance, the secondary diagnostics, and the inference discipline.

4.1 Headline: the discrete-tercile mid-IO concentration

Re-estimate the headline three-way regression (Equation (1), Section 3) separately within each within-stock IO tercile of the stable-HQ subsample. Table 2 reports the result under persistent and time-varying IO.

The mid-IO coefficient is the only tercile with state-clustered $|t| > 2$ on both IO measures and the only tercile clearing the 10-percent wild-cluster-bootstrap threshold (individual $p = 0.064$ persistent, $p = 0.058$ time-varying). A Romano-Wolf step-down family-wise correction across the three documented patterns (mid-IO concentration, within-retail wrong sign, continuous-margin quadratic) returns a family-wise $p = 0.077$ bounded by the discrete-margin tests. The Romano-Wolf $p = 0.077$ is the conservative bound from the original family composition (in which the third pattern, the continuous quadratic, was not significant); recomputation with the current quadratic ($t = 2.28$, $p = 0.017$) would tighten the family-wise p, not loosen it. The family was pre-specified before the functional-form-robustness grid was run. The continuous-margin quadratic ($p = 0.017$, Section 4.2) is a within-pattern rendering of the discrete mid-IO concentration rather than an independent claim, so I report it inside the family. Section 6.1 disciplines the quadratic's five-percent clearance against a parametric grid.

The discrete-tercile mid-IO contrast does not depend on a chosen functional form for IO: it discretizes the within-stock IO distribution into three buckets and reports the coefficient on each. It is the load-bearing characterization of the headline empirical fact.

4.2 Quadratic-IO corroboration of the U-shape

A continuous-margin specification corroborates the non-monotone shape. Augmenting the headline three-way with a quadratic-IO interaction on the stable-HQ subsample,

$$r_{i,s,t} = \alpha_s + \tau_t + \gamma_4\,(\text{mom}\cdot\text{IV}\cdot\texttt{literacy\_z}\cdot\text{IO}) + \gamma_5\,(\text{mom}\cdot\text{IV}\cdot\texttt{literacy\_z}\cdot\text{IO}^2) + \boldsymbol{\beta}'\mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}, \tag{2}$$

where $\mathbf{X}$ collects the 21 lower-order interactions. The literacy slope on mom $\cdot$ IV as a function of IO is $s(\text{IO}) = \gamma_3 + \gamma_4\,\text{IO} + \gamma_5\,\text{IO}^2$, with an interior minimum at $\text{IO}^* = -\gamma_4/(2\gamma_5)$ when $\gamma_5 > 0$. Table 3 reports the result.

The quadratic-IO coefficient $\hat{\gamma}_5 = +0.0151$ on the persistent measure clears wild-cluster bootstrap $p = 0.017$, the implied trough is $\text{IO}^* = 0.344$, and the discrete-tercile mid-IO mean IO share of 0.495 contains the interior trough. The U-shape characterization is therefore consistent across the discrete-tercile contrast and the quadratic specification. The five-percent inferential statement, however, rests on the quadratic functional form: Section 6.1 reports that the same data deliver $p = 0.75$ under a log-IO transformation, $p = 0.24$ under a kink-IO specification at the empirical minimum, and a flat 20-bin conditional-mean diagnostic. The discrete-tercile $p = 0.064$ (individual) and $p = 0.077$ (family-wise) is the inferential statement that does not depend on the parametric form.

4.3 Relocator decomposition: the stable-HQ mid/low ratio is partly sample-selection

The stable-HQ mid/low ratio of 2.83 is larger than the corresponding full-panel ratio of 1.09. Table 4 decomposes the change.

Relocators are nearly twice as concentrated in the low-IO tercile (24.6 percent) as in the other two terciles ($\approx$ 13.7–13.8 percent), and the relocator-only low-IO coefficient is zero. Dropping relocators therefore attenuates the low-IO coefficient (from -0.0170 to -0.0113, a 34-percent reduction) and grows the mid-IO coefficient (from -0.0185 to -0.0321, a 73-percent increase). About one-third of the stable-HQ mid/low gap comes from low-IO attenuation under the relocator drop and about two-thirds from mid-IO growth. The mid-IO peak on the stable-HQ subsample is therefore jointly a mid-IO peak and a low-IO suppression: the underlying within-stock contrast is smaller than the stable-HQ ratio of 2.83 implies. The relocator characteristics (IA G; smaller, higher-IV, lower-return firms) are consistent with a sample-selection contribution to the stable-HQ headline; the magnitude reading is engaged in Section 4.6.

4.4 Pre/post-2015 split: the curvature is stable, the trough drifts

Table 5 reports the pre/post-2015 split on both the discrete-tercile and the continuous-margin quadratic specifications.

The discrete-tercile mid-IO coefficient is directionally preserved (the most-negative tercile in all four cells of Panel A) but neither half is individually significant under wild-cluster bootstrap ($p = 0.24$–0.40). The quadratic curvature $\hat{\gamma}_5$ is constant across the split (+0.0152 pre-2015, +0.0153 post-2015), while the linear-IO four-way magnitude grows and the implied trough IO* drifts from 0.260 to 0.402 as the state-FE-controlled median IO share rises from 0.59 to 0.67. A formal Chow test of $\gamma_5^{\text{pre}} = \gamma_5^{\text{post}}$ on the stable-HQ persistent-IO quadratic-IO four-way ($N = 518{,}962$, 51 states) fails to reject decisively: Wald $\chi^2(1)\, p = 0.997$, $F(1, 518636)\, p = 0.995$, state-clustered contrast $t = -0.003$. The trough is at a fixed relative location within a drifting IO distribution rather than at a fixed absolute IO share. Section 5.2 engages the underpowered Chow diagnostic and the trough-location drift.

4.5 Within-retail wrong sign

On the stable-HQ subsample restricted to the low-IO tercile, the literacy gradient runs the opposite direction of the naive sophistication reading. Splitting at the within-tercile state-mean literacy median, the low-literacy-minus-high-literacy difference in the three-way is +0.0033 persistent (state-clustered

$t = +2.80$, wild-cluster bootstrap $p = 0.024$, 50 states) and $+0.0029$ time-varying ($t = +2.00$, $p = 0.093$, 51 states). The within-mid-IO IV-decile decomposition decomposes to approximately zero across deciles; the persistent-IO joint test of decile equality fails to reject ($p = 0.94$). Within the low-IO tercile itself, a within-month NYSE IV decile decomposition of the same low-literacy-minus-high-literacy difference (IA C) is similarly diffuse: five deciles return positive differences and five return negative; only decile 8 is individually significant; the observation-weighted average is $+0.0002$. The $+0.0033$ DIFF is fully attributable to observation-weighting and decomposes to approximately $+0.0002$ across IV deciles, indistinguishable from zero. The sign reversal is itself an aggregation artifact and provides no positive evidence against the naive sophistication reading; it does not separate within-family candidates.

4.6 Nested baseline: the seed's specific aggregate prediction collapses under PIT

The seed's specific within-channel prediction is a negative aggregate three-way on the full panel and a negative within-stock retail-versus-institutional difference. Table 6 reports both objects under the static-snapshot literacy merge the seed used and under the PIT EDGAR 10-K-header literacy reassignment.

On the full panel under the seed's static-snapshot merge, the aggregate three-way is -0.0117 at marginal significance ($p = 0.078$). Under the PIT reassignment, the aggregate collapses to -0.0003 ($p = 0.517$), approximately 39-fold smaller, and the within-stock retail-versus-institutional difference moves from -0.0211 to $+0.00029$ (sign-flipped). The seed's specific aggregate prediction is an artifact of the contamination introduced by pairing relocator firm-months with the literacy of their static-snapshot headquarters state. Panel B records the corresponding mid-IO coefficient on the PIT full panel: a factor of ≈ 26 smaller ($-0.0321 \rightarrow -0.00121$) than the stable-HQ static headline, with the directional pattern preserved (mid-IO is the only tercile with state-clustered $|t| \geq 2$ on both IO measures) but the magnitude collapsed. The PIT magnitude collapse is the binding population-scope limitation: the headline holds on the stable-HQ subsample where the local-investor unit of analysis is held fixed; on the corrected full panel the pattern is roughly twenty-six times smaller. Two readings of the difference compete on the discriminating margins this design supports: a mechanism-operates-most-strongly-on-stable-HQ reading and a sample-selection-toward-where-the-result-is-strongest reading. The relocator characteristics in IA G lean toward the sample-selection reading; the mechanism-strength reading of the magnitude collapse is not defendable on current evidence.

4.7 Variance-channel diagnostic and college horse-race

A purely statistical reading would attribute the mid-IO concentration to greater identifying variation at the mid tercile. Within-tercile variances of the focal triple mom $\cdot$ IV $\cdot$ `literacy_z` on the stable-HQ subsample are 1.664 (low-IO), 0.673 (mid-IO), and 0.434 (high-IO); high-IO has the lowest dispersion of the three, with mid-IO sitting between low-IO (highest variance) and high-IO (lowest), and Bartlett's equal-variance test rejects strongly. The same ordering survives state-month demeaning. Mid-IO has lower identifying variation than low-IO yet yields a larger slope, so the variance-channel ruling-out applies to the low-IO comparator only; the mid-vs-high ranking is consistent with the variance ordering and is not the discriminating margin.

State financial literacy correlates with state college attainment ($\rho = 0.62$). A horse-race within mid-IO returns $\hat{\gamma}_{\text{lit}} = -0.0350$ ($t = -2.97$) and $\hat{\gamma}_{\text{college}} = +0.00076$ ($t = +0.65$): the literacy three-way sharpens relative to the literacy-only mid-IO baseline of -0.0321, and the college three-way is opposite-signed and indistinguishable from zero. On the stable-HQ aggregate, $\hat{\gamma}_{\text{lit}} = -0.0122$ ($t = -1.65$) and $\hat{\gamma}_{\text{college}} = -0.00051$ ($t = -0.63$). Literacy does not proxy college attainment within mid-IO.

5 Mechanism

5.1 What the data rule out

A generic state-level moderator hits all headquarters-state stocks alike and cannot produce a within-stock tercile-level concentration; the within-stock IO-tercile contrast of Section 4.1 is therefore the family-level discrimination this design supports. The discrimination scope is generic state-level confounds (business cycles, brokerage-participation density, state demographic and industrial mix). Ownership-interactive state-level confounds—a state-level moderator that itself operates differently across IO terciles—remain an open class. The variance-channel diagnostic and the college horse-race (Section 4.7) rule out the simplest statistical and education-proxy alternatives against the low-IO comparator; the HHI alternative (IA M) shows the discrimination is about ownership share, not generic ownership concentration.

The canonical arbitrage-capacity microfoundation is refuted within mid-IO. Within the mid-IO tercile of the stable-HQ subsample (N = 191,589, 2,006 firms), I decompose the headline three-way by within-month NYSE IV decile. The canonical arbitrage-capacity microfoundation puts the strongest moderation at high-IV stocks: arbitrage costs scale with idiosyncratic risk, so literacy-modulated retail demand displaces less mispricing where arbitrage is cheap and more where it is expensive. Table 7 reports the decomposition.

The IV-decile joint test fails to reject decile equality ($\chi^2 = 3.59$, $p = 0.94$); decile 10, where the canonical arbitrage-capacity reading predicts the strongest moderation, returns $\hat{\gamma} \approx 0$. The data rule out the canonical arbitrage-capacity microfoundation within mid-IO. (Decile 6 is individually significant; under time-varying IO decile equality also fails to reject, $\chi^2 = 12.59$, $p = 0.18$.)

The pure-retail margin returns a wrong-signed but artifact-driven null. The within-low-IO literacy gradient runs the opposite direction of the naive sophistication reading (Section 4.5): the low-literacy-minus-high-literacy difference is +0.0033 persistent ($p = 0.024$). The within-retail wrong sign is fully attributable to observation-weighting (see Section 4.5 and IA C). Because the sign reversal is itself an aggregation artifact, the within-retail margin fails to provide positive evidence either for or against the naive sophistication reading; it does not separate within-family candidates.

5.2 The Chow diagnostic on curvature stability is underpowered

Beneath the mid-IO concentration headline, the pre/post-2015 split returns a Chow $\chi^2(1)\, p = 0.997$ that is consistent with equality of $\hat{\gamma}_5$ across the 2015 break but is an underpowered diagnostic, not positive structural evidence. The wild-cluster bootstrap $p = 0.999$ is mechanical for a t-statistic this close to zero (state-clustered contrast $t = -0.003$); γ_5 itself is not individually significant on either half ($t = 1.49$ pre, $t = 1.76$ post); and the contrast standard error is approximately ninety percent of $\hat{\gamma}_5$ itself. The test cannot distinguish stability from absence of evidence.

With that calibration fixed: $\hat{\gamma}_5^{\text{pre}} = +0.0152$ (state-clustered $t = +1.49$) and $\hat{\gamma}_5^{\text{post}} = +0.0153$ (state-clustered $t = +1.76$), with a contrast of -4.3×10^{-5} (state-clustered $t = -0.003$). A formal Chow test of equality on the stable-HQ persistent-IO four-way (N = 518,962 stacked observations, 217,207 pre + 301,755 post, 51 states) fails to reject: Wald $\chi^2(1)\, p = 0.997$ (primary), $F(1, 518636)\, p = 0.995$, and a state-clustered wild-cluster bootstrap with the null imposed and Webb six-point weights ($B = 999$) returns $p = 0.999$ corroborating the asymptotic test. The curvature shape—not its location—is consistent with equality across the break. The trough IO* drifts from 0.26 to 0.40 and the linear-IO four-way magnitude grows, while the curvature $\hat{\gamma}_5$ holds nearly invariant.

The reading the data support is a stable relative location of the trough within a drifting IO distribution. The curvature shape is a property of where literacy moderation lies in the ownership distribution; the absolute IO share at which the moderation peaks does not survive the 2015 secular shift in median IO. The pattern is consistent with a residual non-compositional component, though the test lacks power to distinguish it from a slowly-drifting compositional curvature. I do not lean on the stability claim as identification of a specific within-family microfoundation.

5.3 Within-family readings: candidates the design does not separate

After the within-mid-IO IV-decile refutation of the canonical arbitrage-capacity reading, compositional heterogeneity is the most direct family-internal description of what remains. The mid-IO tercile mechanically carries the largest within-tercile dispersion in actual ownership composition and is likely to be more heterogeneous on other channel-relevant dimensions (firm size, industry, retail-investor age, brokerage density) than the pure terciles. The within-mid-IO IV-decile diffusion on persistent IO is consistent with this reading. The Chow diagnostic on $\hat{\gamma}_5$ documented above is consistent with a residual non-compositional component but is underpowered (Section 5.2); the data do not isolate it. A literacy-correlated local-bias-intensity reading and a mixed-clientele reading are observationally similar to compositional heterogeneity on the discriminating margins this design supports; brokerage-level data and a within-mid-IO sort on within-firm IO dispersion are the next-paper discriminators not available here. A natural family-internal reading referees would anticipate (an arbitrage-capacity Goldilocks with $\mathrm{IO}(1-\mathrm{IO})$ peaking at $\mathrm{IO} = 0.5$) is empirically refuted: the continuous-margin trough is at $\mathrm{IO}^* \approx 0.30$ to 0.34 rather than 0.5 (the three reported sample-and-IO-measure combinations return $\mathrm{IO}^* \in \{0.300, 0.328, 0.344\}$; Table 3), and the within-mid-IO IV-decile decomposition refutes the high-IV-arbitrage microfoundation.

5.4 Interpretive schematic and the posited reduced form

Table 8 is an interpretive schematic, not a causal directed acyclic graph; absent edges are not claimed as tested exclusions.

I posit three reduced-form relationships. Each is a stated empirical pattern, not a derived proposition.

Posit 1 (headline discrete-tercile mid-IO concentration). Stratifying the three-way by within-stock IO tercile,

$$\gamma_{\mathrm{IO,mid}} < 0, \quad |\gamma_{\mathrm{IO,mid}}| > |\gamma_{\mathrm{IO,low}}|, \quad |\gamma_{\mathrm{IO,mid}}| > |\gamma_{\mathrm{IO,high}}|. \tag{3}$$

Estimate (stable-HQ persistent): $\hat{\gamma}_{\mathrm{IO,mid}} = -0.0321$ (individual wcb $p = 0.064$; family-wise Romano-Wolf $p = 0.077$).

Posit 2 (parametric corroborating U-shape; form-fragile). Augmenting the headline three-way with a quadratic-IO interaction (Equation 2), the literacy slope $s(\mathrm{IO}) = \gamma_3 + \gamma_4\,\mathrm{IO} + \gamma_5\,\mathrm{IO}^2$ has an interior trough $\mathrm{IO}^* \in (0,1)$ with $\gamma_5 > 0$. Estimate (stable-HQ persistent): $\hat{\gamma}_5 = +0.0151$ ($p = 0.017$); $\mathrm{IO}^* = 0.344$. The five-percent clearance is parametric and does not survive log-IO or kink-IO alternatives (Section 6.1); the curvature is consistent with equality across the 2015 break under an underpowered Chow diagnostic (Section 5.2).

Posit 3 (within-mid-IO IV-decile diffusion bounds canonical microfoundation). Decomposing the headline three-way within mid-IO by within-month NYSE IV decile, the within-mid-IO IV-decile coefficients are diffuse on the persistent IO measure (joint χ^2 fails to reject equal decile coefficients; $\hat{\gamma}_{d=10} \approx 0$). This bounds the canonical arbitrage-capacity microfoundation.

The seed's specific aggregate and within-stock retail-versus-institutional predictions are stated as the nested baseline the corrected sample refutes; they are not posited.

6 Robustness

This section reports the primary robustness legs: functional-form sensitivity of the quadratic-IO clearance, the classical errors-in-variables (EIV) correction, the wave-anchor and 2009-vintage forward-fill checks, and the inference discipline. The Herfindahl-Hirschman alternative, the Korniotis-Kumar business-cycle overlap, the difference-in-differences secondary design, and the full inference battery are in the Internet Appendix (Sections M, O, T, I).

6.1 Functional-form robustness of the quadratic-IO U-shape

The quadratic-IO five-percent clearance ($\hat{\gamma}_5 = +0.0151$, $p = 0.017$; Section 4.2) rests on the quadratic functional form. I test whether the non-monotone shape survives in alternative parametric specifications and in a fully non-parametric twenty-bin diagnostic on the stable-HQ persistent-IO subsample ($N = 518{,}962$).

The cubic-IO specification has near-collinear focal regressors (variance inflation factors 64, 340, 131); individual cubic coefficients are not separately identifiable, and the row is reported for completeness only. Log-IO returns a coefficient near zero ($p = 0.75$). Kink-IO at the empirical minimum delivers neither a kink coefficient nor a linear term that is statistically separable from zero. The 20-bin diagnostic shows bin-level conditional means within ± 0.003 of zero across the IO range, with no visible monotonic curvature. The 5-percent statistical clearance of the U-shape rests on the quadratic functional form alone; the discrete-tercile mid-IO concentration (Table 2) is functional-form-free and remains the headline characterization.

6.2 Classical errors-in-variables correction

State NFCS literacy estimates are noisy: median Kish effective sample size is ≈ 470 respondents per state-wave. The classical errors-in-variables (EIV) correction divides the slope by the reliability ratio $\lambda \in (0, 1]$, computed from the cross-state variance of state-wave estimates against their mean state-wave sampling variance.

The firm-month-weighted reliability ratio is $\lambda = 0.420$. The EIV-corrected mid-IO coefficient is $\hat{\beta}_{\text{eiv}} = -0.03206/0.4203 = -0.0763$, the direction classical EIV theory predicts. The EIV-corrected number shares the baseline's $t = -2.63$ by construction and is the appropriate economic-magnitude anchor (Section 7); the baseline and EIV-corrected coefficients form one magnitude range on one variation, not two pieces of evidence.

6.3 Wave-anchor and 2009-vintage forward-fill checks

Restricting to the five NFCS wave-anchor years only (2009, 2012, 2015, 2018, 2021; N = 174,226 firm-months, 62,032 in mid-IO), the persistent mid-IO three-way is $\hat{\gamma}_{\text{mid}} = -0.070$ ($t = -2.09$): sharper, not weaker, than the interpolated monthly headline. The headline is not an artifact of forward-fill. Fixing each firm's IO tercile to its 2009 full-year-mean IO assignment (only ≈ 5 percent of firms drift under the persistent IO measure) attenuates the mid/low ratio from 2.83 to 2.09 but preserves the mid-IO concentration. The secular-IO-shift reading of the concentration is bounded.

6.4 Inference discipline

One-way state clustering on the literacy three-way is the binding inference dimension: 51 state clusters are the natural support of the focal variable. The state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed is the appropriate small-cluster procedure. The two-way Cameron-Gelbach-Miller [Cameron et al., 2011] $t = -5.32$ on the static-snapshot full-panel aggregate over-states the evidence relative to the wild-cluster bootstrap $p = 0.078$; I report the wild-cluster bootstrap as the binding statement throughout and show the two-way Cameron-Gelbach-Miller statistic in IA I. Within-tercile state-cluster counts (50 for mid-IO persistent, 49 for mid-IO time-varying; 29–41 for within-mid-IO IV deciles) lie in the range where Webb six-point weights with null imposed maintain test size close to nominal.

7 Discussion

7.1 What the characterization changes

The characterization positions a state-level investor characteristic in two literatures. In household finance, financial literacy has been studied as a predictor of portfolio quality, market participation, and financial resilience [Lusardi and Mitchell, 2014, Calvet et al., 2007, Grinblatt et al., 2012]; I position state NFCS literacy as a candidate cross-sectional asset-pricing moderator on a specific anomaly interaction with a non-monotone IO dependence. In the local-return-predictability program built on Korniotis and Kumar [2013], the contribution adds a literacy moderator alongside the business-cycle channel, distinct but partly overlapping (20–23 percent attenuation under business-cycle controls). Relative positioning to Hillert et al. [2014] and Korniotis et al. [2020] is in the Introduction.

The mid-IO three-way coefficient on the stable-HQ subsample implies a literacy-modulated return differential of -7% annualized at a 0.187-standard-deviation swing in the standardized mom $\cdot$ IV $\cdot$ `literacy_z` composite. The classical-EIV-corrected coefficient $\hat{\beta}_{\text{eiv}} = -0.0763$ implies -16% at the same swing. The relevant economic magnitude is the bracket -7% to -16%, not a point estimate; the EIV-corrected end is the theoretically principled anchor with $t \in [-2.4, -2.8]$ when λ is bootstrapped over the five NFCS waves.

The reliability ratio is estimated; a delta-method standard error accounting for wave-bootstrap uncertainty in λ would be modestly larger and is not reported. The bracket is one magnitude range on one variation rather than two independent pieces of evidence. It overlaps the ≈ 5 percent annualized state-business-cycle strategy of Korniotis and Kumar [2013], with which the literacy moderator is partly correlated.

7.2 Limitations

Population scope. On the corrected PIT full panel the mid-IO coefficient shrinks by a factor of ≈ 26; the mechanism-strength reading of the collapse is not defendable on current evidence (see Section 4.6). The relocator characteristics (IA G) lean toward the sample-selection reading: relocators are smaller, higher-IV, lower-IO, lower-return, and from slightly lower-literacy states. About one-third of the stable-HQ mid/low gap comes from low-IO attenuation under the relocator drop and about two-thirds from mid-IO growth. I do not run a pre/post-relocation event study on the 1,366 dropped relocators; the stable-HQ restriction is motivated by but not tested against the community-tie argument.

Within-family interpretation. The within-family interpretation is bounded as in Section 5.1: compositional heterogeneity is the design-internal description; the design does not separate it from literacy-correlated local-bias intensity, a mixed-clientele reading, or a no-microfoundation null ($p = 0.94$

persistent, $p = 0.18$ time-varying for the within-mid-IO IV-decile equality test). Brokerage-level data and a within-mid-IO sort on within-firm IO dispersion are the next-paper discriminators not available here. The mid-IO coefficient growth from -0.0185 on the full panel to -0.0321 on stable-HQ is a joint sample-composition and reweighting effect; I do not decompose it firm-by-firm.

Functional-form fragility of the U-shape. The continuous-margin quadratic $\hat{\gamma}_5 = +0.0151$ ($p = 0.017$) does not survive log-IO ($p = 0.75$), kink-IO ($p = 0.24$), or a flat 20-bin diagnostic (Section 6.1). The trough-location estimate $\mathrm{IO}^* \approx 0.34$ is correspondingly parametric. The discrete-tercile mid-IO concentration is the headline functional-form-free characterization; the U-shape is its parametric corroboration. The Chow $\chi^2(1)\, p = 0.997$ on $\hat{\gamma}_5^{\mathrm{pre}} = \hat{\gamma}_5^{\mathrm{post}}$ is consistent with equality of the curvature across the 2015 break but does not constitute positive structural evidence: the contrast standard error is approximately ninety percent of $\hat{\gamma}_5$ itself (Section 5.2).

Modern-DiD diagnostics. I do not run the Callaway and Sant'Anna [2021], Sun and Abraham [2021], Goodman-Bacon [2021], Rambachan and Roth [2023] diagnostic stack the pre-registered design specified (IA T for the rationale, the underpowering disclosure, and the pre-registered interpretation-rule departure on the realized $+0.0731$ point estimate).

Ownership-interactive state-level confound class. The within-stock contrast does not discriminate against a state-level moderator that itself operates differently across IO terciles. The canonical instance is state-resident public-pension-fund home-state ownership, where state pensions concentrate holdings in mid-IO firms with mechanical state-level co-movement. A keyword filter of the 13F manager universe identifies 21 distinct state-pension-fund managers concentrated in roughly seven states with active state-pension systems (the modal state-pension state has one or two managers; 44 of 51 states have none in the filter). Even with the 13F home-state-pension-resident control complete, the construction would be quantitatively thin because home-state-pension residence is collinear with the state fixed effect for ≈ 40 of 51 states. State-employee 401(k) custodians and state-resident-brokerage concentration are the closest extensions of this class; I do not identify a separating test on current data. Closing the class would require a CIK-by-CIK state-pension-resident flag matched against the 13F panel; that data source is not built here.

8 Conclusion

This paper characterizes where state NFCS Big-Five financial literacy moderates the momentum-IV interaction in U.S. equities, 2009–2023. On the stable-headquarters subsample, the three-way concentrates in the mid-tercile of within-stock institutional ownership ($\hat{\gamma}_{\text{mid}} = -0.0321$, state-clustered $t =$

-2.63, wild-cluster bootstrap $p = 0.064$; family-wise $p = 0.077$). The discrete-tercile contrast is functional-form-free; a continuous-margin quadratic corroborates a U-shape with interior trough $\text{IO}^* \approx 0.34$, but the five-percent statistical clearance is parametric. Literacy dominates a state-college-attainment horse-race within mid-IO ($\hat{\gamma}_{\text{lit}} = -0.0350$ vs. $\hat{\gamma}_{\text{college}} = +0.00076$). The quadratic-IO curvature $\hat{\gamma}_5$ is consistent with equality across the 2015 break (Chow $\chi^2(1)\, p = 0.997$), even as the trough location IO^* drifts from 0.26 to 0.40; the diagnostic is underpowered (contrast SE $\approx 90\%$ of $\hat{\gamma}_5$; Section 5.2). Within mid-IO, the IV-decile decomposition fails to reject equality across deciles, ruling out the canonical limits-to-arbitrage arbitrage-capacity microfoundation while leaving compositional heterogeneity and observationally-similar non-microfounded readings as candidates the design does not separate. The stable-HQ mid/low ratio of 2.83 is partly a sample-selection effect under the EDGAR relocator drop; the prior aggregate-three-way prediction does not survive the point-in-time EDGAR headquarters correction (the mid-IO coefficient collapses 26-fold and the aggregate three-way collapses approximately 39-fold). The corrected sample refutes the prior aggregate prediction; the documented mid-IO concentration on stable-HQ replaces it as the contribution.

A Variable construction detail

Extended robustness material is in the Internet Appendix: the Herfindahl-Hirschman alternative (IA M), the Korniotis-Kumar business-cycle block (IA O), the difference-in-differences secondary design (IA T), the full PIT literacy reassignment (IA A), the relocator-vs-stable-HQ characteristics comparison (IA G), the Thomson Reuters s34 panel construction (IA P), and alternative literacy constructions (IA L).

NFCS Big Five share-correct score. The state financial-literacy measure is the share of NFCS respondents in a state-wave who answer the FINRA Big Five literacy questions correctly. Correct-answer keys from the FINRA codebook: M6=1, M7=3, M8=2, M9=2, M10=1. An earlier build coded all five questions with a single key; the corrected keys are used throughout, and the Table 1 state-literacy rankings are post-correction. State-wave scores are forward-filled to monthly frequency between waves, then z-scored within calendar month to form `literacy_z`.

Idiosyncratic volatility. Standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window ending in the observation month; z-scored within calendar month before entering the interaction terms.

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Table 1: Descriptive statistics and state literacy rankings

Panel A: Distribution of focal variables (full panel)

Variable	Mean	SD	p05	p50	p95
Monthly return	0.013	0.192	−0.220	0.005	0.253
Momentum (`mom_12_2`)	0.107	0.828	−0.678	0.034	1.021
Idiosyncratic volatility	0.027	0.022	0.007	0.020	0.070
Market equity ($ billions)	7.04	42.1	0.017	0.596	26.19
NFCS Big Five share correct	0.244	0.028	0.200	0.247	0.286

Panel B: Panels and merge conventions

Panel	Firm-months	Firms
Full panel (static-snapshot HQ, baseline literacy merge)	623,896	7,447
Full panel, PIT literacy reassignment	623,814	7,447
Stable-HQ subsample (no in-sample EDGAR relocation)	519,090	6,081
with persistent IO measure	518,962	6,018
with time-varying IO measure	493,889	5,877

Panel C: State literacy rankings and NFCS wave summary

Highest-literacy states	MN, HI, NH, DC, CT
Lowest-literacy states	MS, OH, AR, TX, WV
NFCS wave means (2009, 2012, 2015, 2018, 2021)	0.258, 0.254, 0.255, 0.229, 0.222
Median within-state rank correlation across waves	0.71 (Spearman)

Notes. Idiosyncratic volatility is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. Market equity is price times shares outstanding from CRSP, in billions of dollars. The NFCS Big Five share-correct score is the state-wave share of respondents answering the FINRA Big Five financial-literacy questions correctly. The within-state rank correlation across the five NFCS waves is the median Spearman correlation of state-wave ranks across consecutive waves (a measurement-stability statistic for the state literacy ordering).

Table 2: Mid-IO concentration: three-way coefficient by IO tercile, stable-HQ subsample

Panel A: Persistent IO measure

Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.170	−0.0113	−1.04	0.296	144,989
IO2 (mid)	**0.495**	**−0.0321**	**−2.63**	**0.064**	191,589
IO3 (high; institutional)	0.704	−0.0050	−0.31	0.760	182,384

Panel B: Time-varying IO measure

Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.174	−0.0087	−0.92	0.337	141,129
IO2 (mid)	**0.504**	**−0.0265**	**−2.28**	**0.058**	184,881
IO3 (high; institutional)	0.711	−0.0044	−0.25	0.789	167,879

Notes. Each row reports the three-way coefficient $\hat{\gamma}$ on mom $\times$ IV $\times$ `literacy_z` estimated within the IO tercile, on the stable-HQ subsample under static-snapshot literacy. The persistent measure is each firm's time-mean institutional-ownership share; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier. Wild-cluster bootstrap uses Webb six-point weights [Cameron et al., 2008] with $B = 4{,}999$ and null imposed. Sample: January 2009 through December 2023. Under the 2009-vintage tercile assignment (each firm's tercile fixed at its 2009-full-year-mean IO), ≈ 5 percent of firms drift terciles and the mid/low ratio attenuates from 2.83 to 2.09 (Section 6.3).

Table 3: Quadratic-IO four-way: corroborating U-shape, stable-HQ

Sample / IO measure	$\hat{\gamma}_4$ (linear)	state-t	wcb p	$\hat{\gamma}_5$ (quadratic)	state-t	wcb p
Stable-HQ persistent (headline)	**−0.0104**	**−2.02**	**0.033**	**+0.0151**	**+2.28**	**0.017**
Stable-HQ time-varying	−0.0083	−1.71	0.106	+0.0139	+2.21	0.062
Full panel persistent (reference)	−0.0068	−1.71	0.069	+0.0104	+2.02	0.037
Implied interior trough $\text{IO}^* = -\gamma_4/(2\gamma_5)$:						
Stable-HQ persistent				$\text{IO}^* = 0.344$		
Stable-HQ time-varying				$\text{IO}^* = 0.300$		
Full panel persistent				$\text{IO}^* = 0.328$		

Notes. The coefficients are on the standardized four-way interactions mom $\times$ IV $\times$ `literacy_z` $\times$ IO and mom $\times$ IV $\times$ `literacy_z` $\times$ IO^2 entered jointly with all 21 lower-order terms. State and month fixed effects absorbed. Wild-cluster bootstrap with Webb six-point weights [Webb, 2014, Cameron et al., 2008], $B = 4{,}999$, null imposed. Sample sizes: stable-HQ persistent $N = 519{,}090$; stable-HQ time-varying $N = 493{,}889$; full-panel persistent $N = 623{,}746$.

Table 4: Relocator decomposition: by-tercile coefficients across sample restrictions

Sample	IO1 (low) $\hat{\gamma}$	IO2 (mid) $\hat{\gamma}$	IO3 (high) $\hat{\gamma}$	mid/lo
Full panel ($N = 623{,}896$)	-0.0170 ($t = -2.05$)	-0.0185 ($t = -2.01$)	$+0.0043$	
Stable-HQ ($N = 519{,}090$)	-0.0113 ($t = -1.04$)	-0.0321 ($t = -2.63$)	-0.0050	
Relocator-only ($N = 104{,}806$, 1,366 firms)	-0.0077 ($t = -0.37$)	-0.0124 ($t = -0.91$)	$+0.0085$	
Relocator share of each tercile on the full panel: IO1 24.6%; IO2 13.7%; IO3 13.8%.				

Notes. The three-way coefficient $\hat{\gamma}$ within each IO tercile (persistent IO) under static-snapshot literacy. State-clustered t-statistics in parentheses. Internet Appendix G reports the relocator-vs-stable-HQ firm-mean characteristics.

Table 5: Pre/post-2015 split: discrete tercile and continuous-margin quadratic, stable-HQ

Panel A: Discrete-tercile mid-IO coefficient

Subsample	Mid-IO $\hat{\gamma}$ (persistent)	wcb p	Mid-IO $\hat{\gamma}$ (time-varying)	wcb p
Pre-2015 (2009–2014)	-0.0389 ($t = -1.87$)	0.236	-0.0351 ($t = -2.33$)	0.228
Post-2015 (2015–2023)	-0.0259 ($t = -1.15$)	0.266	-0.0209 ($t = -0.88$)	0.398

Panel B: Continuous-margin quadratic-IO four-way (persistent IO, stable-HQ)

Subsample	$\hat{\gamma}_4$	wcb p	$\hat{\gamma}_5$	wcb p	IO*
Pre-2015 ($N = 217{,}095$)	-0.0079 ($t = -1.16$)	0.295	$+0.0152$ ($t = +1.49$)	0.171	0.260
Post-2015 ($N = 301{,}995$)	-0.0123 ($t = -1.72$)	0.089	$+0.0153$ ($t = +1.76$)	0.089	0.402
Within-state-FE median IO share: 0.59 pre-2015; 0.67 post-2015.					

Notes. Panel A reports the discrete-tercile mid-IO coefficient on the two halves. Panel B reports the continuous-margin quadratic-IO four-way on the same halves; the trough IO$^* = -\gamma_4/(2\gamma_5)$. The 2009-vintage tercile assignment (Section 6) attenuates the mid/low ratio from 2.83 to 2.09 with ≈ 5 percent of firms drifting.

Table 6: The nested baseline: seed's specific predictions and their PIT collapse

Panel A: Aggregate three-way coefficient

Specification	$\hat{\gamma}$	State-t	wcb p	N
Full panel, static-snapshot literacy (seed's convention)	−0.0117	−2.17	0.078	623,896
Full panel, PIT literacy reassignment	−0.0003	−0.86	0.517	623,814
Stable-HQ subsample, static literacy	−0.0145	−1.94	0.122	519,090

Panel B: Mid-IO coefficient on the PIT full panel (binding population-scope limitation)

Specification	Mid-IO $\hat{\gamma}$	State-t	wcb p
Stable-HQ static, persistent (Table 2)	−0.0321	−2.63	0.064
PIT full-panel, persistent	−0.00121	−2.07	—
PIT full-panel, time-varying	−0.00097	−2.35	—
Magnitude collapse (stable-HQ / PIT)	$\approx 26\times$		

Notes. The static-snapshot merge pairs each firm-month with the literacy of its Compustat `state`; the PIT reassignment re-merges against the firm's then-current 10-K-header state and re-z-scores within month. State-level wild-cluster bootstrap (Webb six-point, $B = 4{,}999$ or 9,999, null imposed). Full PIT diagnostic stack in IA A; PIT mid-IO table in IA E.

Table 7: Within-mid-IO IV-decile decomposition (persistent IO, stable-HQ)

IV decile (within-firm IV percentile)	$\hat{\gamma}$	State-t	N
1 (lowest IV; firm IV mean = −0.87)	−0.039	−0.31	19,430
2	−0.049	−0.37	28,223
3	−0.053	−0.76	25,183
4	−0.024	−0.35	23,345
5	−0.038	−0.68	20,191
6	**−0.090**	**−2.30**	17,853
7	+0.000	−0.00	18,459
8	−0.050	−1.03	16,458
9	−0.028	−1.09	13,880
10 (highest IV; firm IV mean = +1.15)	−0.004	−0.11	8,567
Bottom-3 average	−0.047	—	72,836
Top-3 average	−0.027	—	38,905
Joint test, equal-decile χ^2 (df = 9)	3.59, $p = 0.94$	—	—

Notes. The coefficient $\hat{\gamma}$ is on mom $\times$ IV $\times$ `literacy_z` estimated within the mid-IO (IO2) tercile of the stable-HQ subsample, restricted to the IV decile indicated. Deciles are computed within-firm via `pd.qcut` on firm-mean IV. State-clustered t-statistics. Joint test is a state-cluster-SE-weighted χ^2 of equal decile coefficients. Persistent IO measure.

Table 8: Interpretive schematic of the channel family, the nested baseline, and the documented patterns

Object	**Content**
Channel family (topic)	Local bias $\times$ limits-to-arbitrage $\times$ sophistication-dependent local retail demand on HQ-state firms, transmitted through state NFCS literacy.
Nested baseline (prior aggregate prediction)	Aggregate three-way negative; retail-versus-institutional difference negative. Refuted under PIT (Section 4.6).
Stable-HQ headline population	6,081 firms with no in-sample EDGAR relocation.
Headline (discrete tercile)	$\hat{\gamma}_{\mathrm{mid}} = -0.0321$, $p = 0.064$ individual, family-wise $p = 0.077$. Functional-form-free.
Corroborating quadratic U-shape	$\hat{\gamma}_5 = +0.0151$, $p = 0.017$, $\mathrm{IO}^* = 0.34$. Five-percent clearance is parametric (Section 6.1).
Chow diagnostic on curvature γ_5	$\chi^2(1)\, p = 0.997$; consistent with equality across the 2015 break, but underpowered (contrast SE $\approx 0.9 \times \hat{\gamma}_5$); trough IO^* drifts.
Within-mid-IO IV-decile	Diffuse under persistent IO ($p = 0.94$); rules out canonical arbitrage-capacity microfoundation.
Family-internal candidates (parallel; design does not separate)	Compositional heterogeneity; arbitrage-capacity (refuted on the IV-decile margin); literacy-correlated local-bias intensity; no-channel null (cannot be ruled out).
Population-scope limitation	PIT magnitude collapse $\approx 26\times$ on the corrected full panel.

Notes. Interpretive schematic; not a causal directed acyclic graph.

Table 9: Functional-form grid: the quadratic five-percent clearance does not survive

Specification	Focal term	Coef	State-t	wcb p
Quadratic (corroborating)	γ on IO^2	**+0.0151**	**+2.28**	**0.017**
Cubic	γ on IO^2	−0.733	−1.23	0.268
Cubic	γ on IO^3	+0.798	+1.70	0.133
Log-IO	γ on log(IO)	−0.00243	−0.35	0.748
Kink-IO at 0.30	kink coef	+0.179	+1.19	0.242
Kink-IO at 0.30	linear term	−0.116	−1.11	0.288
20-bin diagnostic	conditional means within ±0.003 of zero	—	—	—

Notes. All on the stable-HQ persistent-IO subsample. Each row reports the focal-basis coefficient with state-clustered t and wild-cluster bootstrap p. The 20-bin diagnostic partitions persistent IO into 20 quantile bins, residualizes the focal triple's contribution to monthly returns on state and month fixed effects within each bin, and reports the within-bin conditional mean of the demeaned composite. Bin-level means range from −0.00382 to +0.00361; full 20-bin table in IA F.

Table 10: Classical errors-in-variables correction of the mid-IO three-way

Specification	$\hat{\beta}$	State-t	Note
Baseline mid-IO (stable-HQ, persistent)	−0.0321	−2.63	Table 2
EIV-corrected ($\boldsymbol{\hat{\beta}_{\text{eiv}} = \hat{\beta}/\lambda}$)	**−0.0763**	**−2.63**	2.4× inflation

Notes. Firm-month-weighted reliability ratio $\lambda = 0.420$ from per-wave λ_w: 0.211 (2009), 0.476 (2012), 0.359 (2015), 0.562 (2018), 0.517 (2021). $\lambda_w = (V_{\text{obs}} - V_{\text{err}})/V_{\text{obs}}$ where $V_{\text{err}} = \hat{p}(1-\hat{p})/n_{\text{eff}}$. The standard error treats λ as known; bootstrapping λ over the five NFCS waves moves the EIV t into $[-2.4, -2.8]$. Full per-wave detail in IA N.